

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- No Cummins® service tools required.

Additional Service Items

- Compressed air
- Hoist.

Preparatory Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

WARNING

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

WARNING

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

CAUTION

Use caution when draining coolant that coolant is not spilled or drained into the bilge area. Do not pump the coolant overboard. If the coolant is not reused, it must be discarded in accordance with local environmental regulations.

⚠ CAUTION ⚠

Zinc plugs expand and can break off during removal. Inspect the zinc plug to make sure it is in one piece. If not, it must be replaced with a new zinc plug, and the broken pieces must be retrieved from the heat exchanger to prevent damage to components downstream in the sea water system.

Note : This procedure applies to marine engines only.

Note : If a check for a turbocharger gasket leak is being performed, do not break the seal between the turbocharger and the exhaust manifold. Remove the two components as one assembly.

- Disconnect the batteries. Refer to Procedure 013-009 in Section 13.
(/qs3/pubsys2/xml/en/procedures/99/99-013-009.html)
- Close the sea water supply valve. See equipment manufacturer service information.
- Drain the sea water from the heat exchanger by removing the zinc plug from the bottom of the heat exchanger. Refer to Procedure 008-053 in Section 8.
(/qs3/pubsys2/xml/en/procedures/41/41-008-053.html)
- Drain the engine coolant. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/100/100-008-018-tr.html)
- Remove the exhaust outlet connection. See equipment manufacturer service information.
- Remove the turbocharger heat shield. See equipment manufacturer service information.
- Remove the coolant expansion tank, if equipped. See equipment manufacturer service information.
- Remove the coolant heat exchanger. Refer to Procedure 008-053 in Section 8.
(/qs3/pubsys2/xml/en/procedures/41/41-008-053.html)

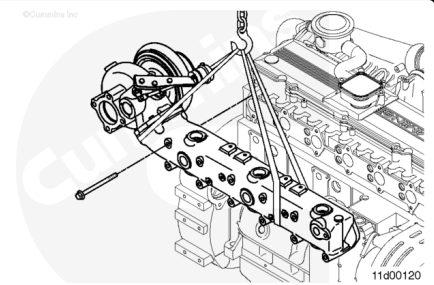
Note : If the turbocharger and exhaust manifold are to be removed as an assembly, then continue on with the following steps. If not, Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/41/41-010-033-tr.html)

- Disconnect the closed crankcase system from the rockerlever cover, if equipped. See equipment manufacturer service information.
- Remove the air crossover from the turbocharger. Refer to Procedure 010-019 in Section 10.
(/qs3/pubsys2/xml/en/procedures/100/100-010-019.html)
- Remove the turbocharger coolant hoses from the turbocharger. See equipment manufacturer service information.
- Remove the turbocharger oil drain line from the turbocharger. Refer to Procedure 010-045 in Section 10. (/qs3/pubsys2/xml/en/procedures/100/100-010-045.html)
- Remove the turbocharger oil supply line from the turbocharger. Refer to Procedure 010-046 in Section 10. (/qs3/pubsys2/xml/en/procedures/100/100-010-046.html)
- Remove the coolant vent line. Refer to Procedure 008-017 in Section 8.
(/qs3/pubsys2/xml/en/procedures/41/41-008-017.html)

Remove

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Remove the coolant supply line.

Loosen two capscrews and leave them in place to hold the weight of the exhaust manifold.

Remove the remaining 10 exhaust manifold mounting capscrews.

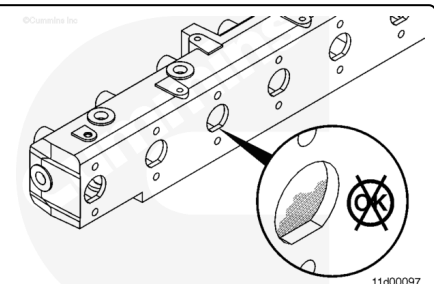
Attach a lifting device to support the weight of the exhaust manifold.

Remove the two remaining mounting capscrews and the exhaust manifold.

Clean and Inspect for Reuse

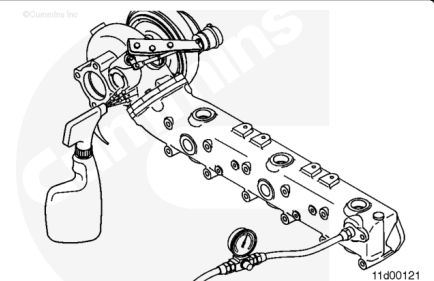
Inspect all the sealing surfaces.

Inspect the inside of the exhaust manifold for possible signs of leakage (rust marks, antifreeze stains, carbon washed away).



⚠ CAUTION ⚠

If the turbocharger was not removed from the exhaust manifold, do not place the turbocharger into water. Damage to the turbocharger will occur if water enters any of the turbocharger openings.



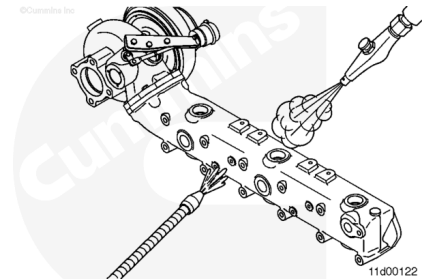
Plug all open coolant passages.

If the turbocharger was **not** removed from the exhaust manifold, pressure test the exhaust manifold and turbocharger as a unit.

Check all sealing surfaces with soapy water for possible leaks.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

Cover all openings of the turbocharger before cleaning or repainting the exhaust manifold. Damage will result if paint enters the turbocharger.

If no leaks are found, clean and repaint the exterior surfaces of the exhaust manifold.

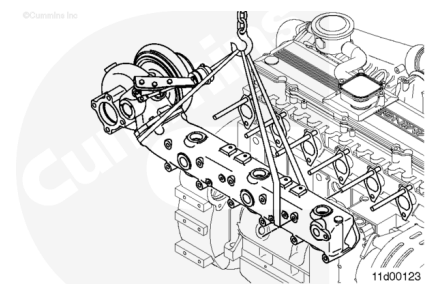
Use solvent to clean the turbocharger and exhaust manifold assembly.

Dry with compressed air.

Install

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Note : The exhaust manifold gaskets are bi-directional and can be installed in either direction.

Install six guide pins into the cylinder head to aid in installing the exhaust manifold assembly.

Install new exhaust manifold gaskets over the studs.

Use a hoist to install the exhaust manifold over the guide studs.

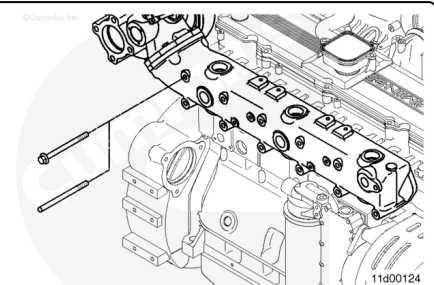
Install the upper six capscrews, but do **not** tighten at this time.

Remove the guide pins one at a time and install the mounting capscrew. Make sure the gaskets stay in place during this process.

Tighten the capscrews in a crisscross pattern from the center out.

Torque Value: 43 n•m [32 ft-lb]

Install the exhaust manifold coolant supply line.



Finishing Steps

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the coolant vent lines. Refer to Procedure 008-017 in Section 8.
(/qs3/pubsys2/xml/en/procedures/41/41-008-017.html)

Note : If the turbocharger and exhaust manifold were removed as an assembly, then continue on with the following steps. If not, Refer to Procedure 010-033 in Section 10. (</qs3/pubsys2/xml/en/procedures/41/41-010-033-tr.html>)

- Install the turbocharger oil supply line to the turbocharger. Refer to Procedure 010-046 in Section 10. (</qs3/pubsys2/xml/en/procedures/100/100-010-046.html>)
- Install the turbocharger oil drain line to the turbocharger. Refer to Procedure 010-045 in Section 10. (</qs3/pubsys2/xml/en/procedures/100/100-010-045.html>)
- Install the turbocharger coolant hoses. See equipment manufacturer service information.
- Install the air crossover to the turbocharger. Refer to Procedure 010-019 in Section 10. (</qs3/pubsys2/xml/en/procedures/100/100-010-019.html>)
- Connect the closed crankcase system, if equipped. See equipment manufacturer service information.
- Install the coolant heat exchanger. Refer to Procedure 008-053 in Section 8. (</qs3/pubsys2/xml/en/procedures/41/41-008-053.html>)
- Install the coolant expansion tank, if equipped. See equipment manufacturer service information.
- Install the turbocharger heat shield. See equipment manufacturer service information.
- Install the exhaust outlet connection. See equipment manufacturer service information.
- Fill the engine with coolant. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/100/100-008-018-tr.html>)
- Open the sea water supply valve(s). See equipment manufacturer service information.
- Connect the batteries. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)

Operate the engine. Check for leaks.